

Unit 4

STRATEGIC INTERACTIONS AND SOCIAL DILEMMAS

OUTLINE

A. Introduction

B. Game theory

C. Types of equilibrium

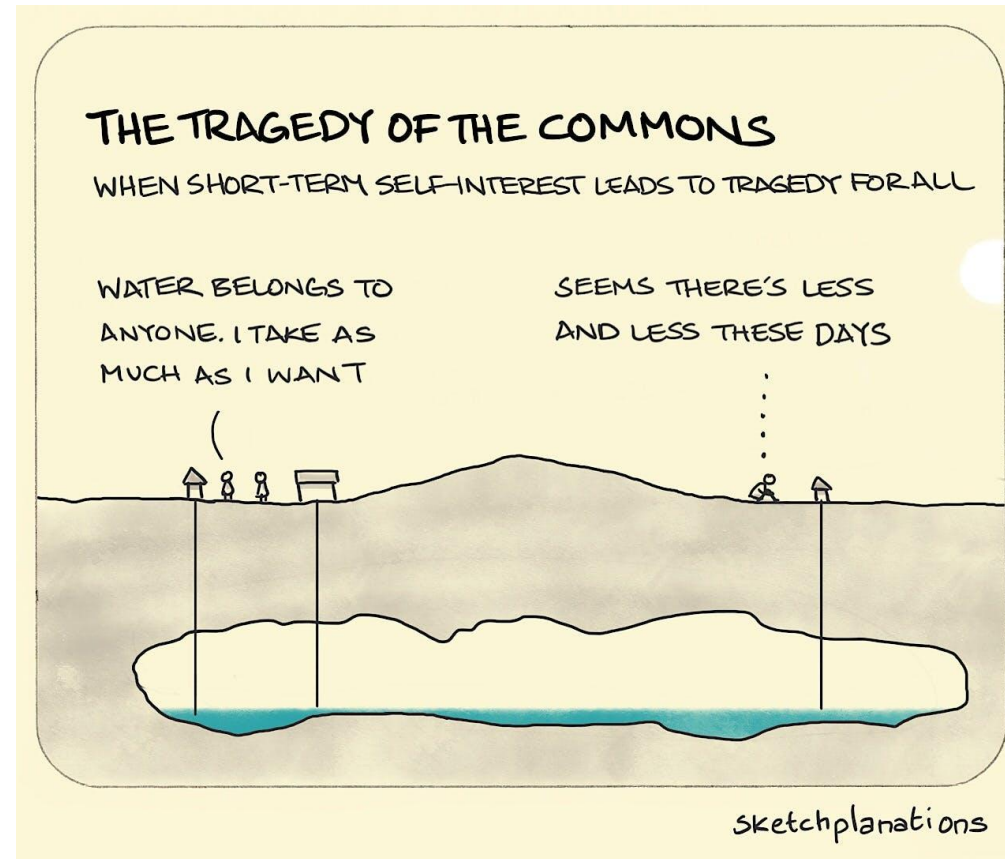
D. Unit 4 Make-up Assignment

A. Introduction

The Context for This Unit

- Individuals choose the best possible outcome given constraints (Unit 3)
- Sometimes the outcome depends on the actions of others as well as your own actions
 - How does that affect decision making?
 - Does this mean that norms and institutions matter?

This Unit



Social interactions can help explain the Tragedy of the Commons and social dilemmas like climate change.

This Unit

- **Social dilemmas** occur when people do not take adequate account of the effects of their actions on others, whether these are positive or negative.

This Unit

Use a **model of social interaction** to explain:

- situations in which there are two or more people,
- and the actions taken by each person affect both their own outcome,
- and other people's outcomes.

B. Game theory

What is game theory?

- **Game theory** is a set of models of **strategic interactions**.
- When people are engaged in a social interaction and are aware of the ways that their actions affect others, and vice versa, we call this a **strategic interaction**.
- A **strategy** is defined as an action (or action plan) that a person may choose while aware of the mutual dependence of the outcomes on their own and others' actions.

What is a game?

A game is a description of a **social interaction**, which specifies:

- *The players*: Who is interacting with whom
- *The feasible strategies*: Which actions are open to the players
- *The order of play*: When the players choose their actions
- *The information*: What each player knows when making their decision
- *The payoffs*: What the outcomes will be for each of the possible combinations of actions.

An example of a simultaneous game

Imagine two farmers, who we will call Anil and Bala.

They face a problem: should they grow rice or cassava? We assume that they have the ability to grow both types of crop, but can only grow one type at a time.

Anil's land is equally suitable for growing rice and cassava. Bala's land is likewise good for producing rice, but less suitable for cassava. They both sell whatever crop they produce in a nearby village market. On market day, if they bring less rice to the market, the price will be higher. Likewise, the price of cassava depends on how much cassava they have grown.

An example of a simultaneous game

		Bala	
		Rice	Cassava
Anil	Rice	Both produce rice There is a glut of rice (low price) There is a shortage of cassava	No market glut High prices for both crops Bala produces the crop for which his land is unsuited
	Cassava	No market glut High prices for both crops	Both produce cassava There is a glut of cassava (low price) There is a shortage of rice Bala produces the crop for which his land is unsuited

		Bala	
		Rice	Cassava
Anil	Rice	Anil gets 4 Bala gets 4	Anil gets 6 Bala gets 3
	Cassava	Anil gets 6 Bala gets 6	Anil gets 5 Bala gets 2

Define the five elements of this game...

- *The players:*
- *The feasible strategies:*
- *The order of play:*
- *The information:*
- *The payoffs:*

Types of Games

Type of Game	Key Feature	Example	Real-World Context	Famous Game
Cooperative vs. Non-cooperative	Binding agreements possible?	Cartel vs. farmers using pesticide	OPEC quotas vs. secret price cuts	Prisoner's Dilemma
Simultaneous vs. Sequential	Moves at same time or in turns	Rock–paper–scissors vs. chess	Price setting vs. market leader/follower	Stackelberg model, Centipede
Zero-sum vs. Non-zero-sum	Gains equal losses?	Poker vs. trade deals	Sports betting vs. climate treaties	Matching Pennies
One-shot vs. Repeated	Played once or many times	One-time price war vs. repeated rivalry	Short-term deal vs. long-term cooperation	Iterated Prisoner's Dilemma
Perfect vs. Imperfect Information	Are moves visible?	Chess vs. poker	Public negotiations vs. hidden markets	Signaling games

C. Types of equilibrium

Finding best responses

		Bala	
		Rice	Cassava
Anil	Rice	4 4	3 6
	Cassava	6 6	2 5

Nash equilibrium

In game theory, a set of strategies in which each player plays a best response to the strategies of the other players is called a **Nash equilibrium**.

In the Nash equilibrium of this game

- Anil chooses Cassava
- Bala chooses Rice

		Bala	
		Rice	Cassava
Anil	Rice	4, 4 ○	3, 6 ●
	Cassava	6, 6 ⊙	2, 5

Dominant strategy equilibrium

If the best response of one player is always the same strategy, regardless of the strategy of the other player, then we say that this is the player's **dominant strategy**.

		Bala	
		Rice	Cassava
Anil	Rice	3, 5	4, 4
	Cassava	6, 6	5, 3

Prisoner's dilemma

Imagine that Anil and Bala are now facing another problem. Each is deciding how to deal with pest insects that destroy the crops they cultivate in their adjacent fields. Each has two feasible strategies:

- The first is to use an inexpensive chemical called Toxic Tide. It kills every insect for miles around. Toxic Tide also leaks into the water supply that they both use.
- The alternative is to use integrated pest control (IPC), in which beneficial insects are introduced to the farm. The beneficial insects eat the pest insects.

Prisoner's dilemma

		Bala	
		IPC	Toxic Tide
Anil	IPC	Beneficial insects spread over both fields, eliminating pests No water contamination	Bala's chemicals spread to Anil's field and kill his beneficial insects Limited water contamination
	Toxic Tide	Anil's chemicals spread to Bala's field and kill his beneficial insects Limited water contamination	Eliminates all pests Heavy water contamination Requires costly filtration system

		Bala	
		IPC	Toxic Tide
Anil	IPC	3	4
	Toxic Tide	1	2

Types of Equilibria

Equilibrium Type	Key Idea	Example	Real-World Context	Famous Game
Nash Equilibrium	No one gains by deviating	Both farmers use pesticide	Countries not reducing CO2	Prisoner's Dilemma
Dominant Strategy	Best choice regardless	"Always use pesticide"	Online auctions → bid true value	Prisoner's Dilemma
Pareto-efficient	No one better off w/out hurting others	Both use IPC	Climate treaties	Tragedy of the Commons
Subgame Perfect	Nash in every stage	Firm A sets price, Firm B reacts	Sequential bargaining	Centipede Game
Mixed Strategy	Players randomize	Rock–paper–scissors	Football penalty kicks	Matching Pennies

D. Unit 4 Make-up Assignment

- Please open Cyber Campus, Week 4, “Unit 4 Make-up Assignment.docx”

Summary

1. Social interactions can be modeled as **games**
 - **Players** choose **best responses** to others' **strategies**
2. **Social dilemmas** e.g. prisoners dilemma can be resolved by social preferences, peer punishment, or binding agreements
 - The rules of the game also matter for outcomes
3. Multiple Nash equilibria can cause coordination problems
 - Economic and political institutions can help achieve socially optimal outcomes

In the next unit

- The role of **institutions** in social allocations
- How to evaluate social outcomes: **efficiency** and **fairness**
- How bargaining power affects the distribution of surplus